STATISTICAL ANALYSIS REPORT

Customer Churn Prediction – Week 3 Machine Learning Project

1. Introduction

This report presents the statistical analysis performed on the processed telecom customer dataset used for the customer churn prediction project. The analysis focuses on the distribution of the target variable, descriptive statistics for key numerical variables, and observed relationships between selected variables.

2. Dataset Overview

The processed dataset contains 7,043 customer records and 32 variables. The dataset includes numerical variables and one-hot encoded categorical variables, with Churn serving as the target variable. The analysis was performed after data preprocessing and feature transformation.

3. Churn Distribution

The target variable is moderately imbalanced. There are 5,174 customers who did not churn, representing 73.46% of the dataset, while 1,869 customers churned, representing 26.54%. This distribution was considered when evaluating the classification models because a model can achieve relatively high accuracy while still failing to identify a substantial proportion of churners.

Churn Status	Count	Percentage
No Churn (0)	5,174	73.46%
Churn (1)	1,869	26.54%

4. Descriptive Statistics

Descriptive statistics were examined for the main numerical variables. The mean, standard deviation, minimum, and maximum provide information about central tendency and spread.

Variable	Mean	Std. Dev.	Minimum	Maximum
SeniorCitizen	0.1621	0.3686	0	1
tenure	32.3711	24.5595	0	72
MonthlyCharges	64.7617	30.0900	18.25	118.75
TotalCharges	2279.7343	2266.7945	0	8684.80

5. Interpretation of Numerical Variables

Tenure has a mean of approximately 32.37 months and a standard deviation of approximately 24.56 months, with a range of 0 to 72 months. MonthlyCharges has a mean of approximately 64.76 and a standard deviation of approximately 30.09, with values from 18.25 to 118.75. TotalCharges has a mean of approximately 2,279.73 and a standard deviation of approximately 2,266.79, ranging from 0 to 8,684.80. SeniorCitizen has a mean of approximately 0.162, corresponding to approximately 16.21% of observations in the senior-citizen category.

6. Correlation Analysis

Correlation analysis was used to examine the strength and direction of linear relationships between selected numerical variables. Correlation measures association and does not establish causation.

Variable 1	Variable 2	Correlation
tenure	TotalCharges	0.8262
MonthlyCharges	TotalCharges	0.6512

7. Interpretation of Correlations

The correlation between tenure and TotalCharges is 0.8262, indicating a strong positive linear association. This is expected because customers who remain with the company for longer periods have more time to accumulate charges.

The correlation between MonthlyCharges and TotalCharges is 0.6512, indicating a moderately strong positive relationship. Customers with higher monthly charges tend to have higher cumulative charges, although the relationship is also influenced by the length of the customer relationship.

These correlations indicate that some numerical predictors contain overlapping information, which should be considered when interpreting model coefficients or feature importance.

8. Statistical Findings Relevant to Modelling

- The target distribution is imbalanced, with churn representing 26.54% of observations.
- Tenure and TotalCharges show a strong positive association, so their overlapping information should be considered during model interpretation.
- MonthlyCharges and TotalCharges also show a meaningful positive association.
- The numerical variables have different scales, supporting scaling for models that are sensitive to feature magnitude.
- Statistical summaries provide descriptive evidence but do not by themselves establish causal relationships with churn.

9. Business Interpretation

The statistical analysis indicates substantial variation in customer relationship duration and charges. The churn distribution also shows that churn is a minority outcome, making it important for the business to focus on identifying churners rather than relying only on overall classification accuracy.

The strong association between tenure and TotalCharges suggests that customer relationship duration is closely linked to cumulative revenue. The relationship between MonthlyCharges and TotalCharges also indicates that charge levels may be relevant signals when analysing customer behaviour.

10. Limitations

The analysis is primarily descriptive and correlational. Reported correlations do not demonstrate causality, and descriptive statistics do not by themselves determine which variables cause churn. Further statistical testing or causal analysis would be required to make causal claims.

11. Conclusion

The statistical analysis provides a quantitative overview of the processed customer dataset and establishes important context for the machine learning analysis. The dataset contains 7,043 observations, with 26.54% churn. Tenure and TotalCharges have a strong positive correlation of 0.8262, while MonthlyCharges and TotalCharges have a correlation of 0.6512. These findings support interpretation of the churn-prediction models and their feature-importance results.